

Variable Bench Power Supply With LCD and Monitor Display



ABHISHEK KUMAR

Presented here is an economical and cost-effective yet efficient variable bench power supply that provides one variable and two fixed supply channels. It provides 1.2V to 25V supply with up to 5-ampere current through one channel, and 5V, 1A and 12V, 1A supplies through the other two channels. It also displays the voltage on the output terminals, and the current and instantaneous power drawn by the load on an on-board LCD display. You can connect this power supply to your computer via the serial port and see the voltage, current and power drawn by the load graphically.

The project has two sections: the main power module having linear voltage regulators with rectification and filtering circuitry for supply generation and regulation, and the other having a microcontroller that is used to sense and display the current and voltage across the variable supply channel.

Block diagram

The block diagram is in two parts, one for each section:

Main power section. Block diagram of the main power section is shown in Fig. 2.

This section is used to derive a regulated DC voltage from the mains 230V AC input. It has four sections. The first section steps down 230V AC input to 20V AC. The second section rectifies the AC component of its input signal. The third section acts as a first-order low-pass filter and also smoothen the signal.

The fourth section for regulation is further divided into three sections, one for each channel. Thus, at the end, we get three channels of the power supply.

Interface and display section. Block



Fig. 1: Author's prototype and display modes

diagram of the interface and display section is shown in Fig. 3. This section senses the current and voltage on the variable supply channel and shows readings on the LCD display and computer monitor.

It consists of a voltage divider section and a series shunt arrangement at the variable supply channel for sensing the voltage across the terminals and current through load, respectively. A non-inverting amplifier is used after the shunt arrangement to amplify the voltage drop across the shunt resistor. The outputs of the voltage divider network and the non-inverting amplifier go to analogue-to-digital converter (ADC) inputs of a microcontroller. The ADC converts this real-time analogue data into digital data so that it can be processed in the microcontroller, interpreted and sent off to the serial interface and display section. Two buttons are provided to act as inputs and a buzzer is used to give audio indications.

Power stage

Fig. 4 shows the circuit of the main power section. It comprises a step-down transformer (X1), four power diodes (D1-D4), adjustable regulator LM338 (IC4), regulator 7812 (IC2),

PARTS LIST

Semiconductors:

- IC1 - ATmega8 microcontroller
- IC2 - 7812, +12V regulator
- IC3, IC6 - 7805, +5V regulator
- IC4 - LM338 adjustable voltage regulator
- IC5 - 741 op-amp
- IC7 - MAX232 RS232 driver
- D1-D4 - MBR750 rectifier diode
- D5-D6 - 1N4001 rectifier diode
- BR1 - 1A bridge rectifier module
- LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, 5% carbon):

- R1 - 240-ohm
- R2 - 2-kilo-ohm
- R3, R9, R10 - 10-kilo-ohm
- R4, R6 - 100-kilo-ohm
- R5 - 1-mega-ohm
- R7 - 70-kilo-ohm
- R8 - 220-ohm
- VR1 - 5-kilo-ohm potentiometer
- VR2 - 10-kilo-ohm preset
- Shunt resistor - 0.3-ohm, 10 watt

Capacitors:

- C1-C4 - 100nF ceramic disk
- C5 - 4700µF, 50V electrolytic
- C6 - 10µF, 40V electrolytic
- C7 - 1000µF, 25V electrolytic
- C8, C9 - 0.1µF ceramic disk
- C10-C13 - 1µF, 16V electrolytic
- C14 - 10µF, 16V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 20V, 6A secondary transformer
- X2 - 230V AC primary to 9V, 500mA secondary transformer
- PZ1 - Piezo buzzer
- S1-S3 - Tactile switch
- DB9 - 9-pin D-type serial COM port connector
- LCD1 - 16x2 LCD module
- CON1 - 4-pin connector
- CON2 - 3-pin connector
- F1 - 6A fuse

regulator 7805 (IC3) and a few other components.

A fuse of 6A rating is connected in series with the primary of the transformer to prevent short circuit. Step-down transformer X1 converts 230V AC to 20V AC and has a current rating of 5 to 6 amperes. The four Schottky power diodes D1-D4 have current rating of 7.5 ampere. G820 or P600 may be used in place of MBR750, or even a single-module bridge rectifier can be used. A 100nF, 50V capacitor is connected in parallel to each of the rectifier diodes (D1 through D4) for smoothening ripples.

LM338 is an adjustable 3-terminal